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DIFFERENTIAL EQUATIONS FOR ENGINEERS

Analytical Solutions and Numerical Algorithms

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Chapter 1

FUNDAMENTALS OF DIFFERENTIAL EQUATIONS: INTRODUCTION

Differential equations are among the most powerful tools in mathematics for describing the physical world. They arise naturally whenever we seek to understand how quantities change over time or space — from the motion of planets to the growth of populations, from the flow of electric current to the dissipation of heat.

This introductory chapter lays the foundation for the entire course. We begin by exploring **applied problems** that lead to differential equations, illustrating how mathematical models are constructed from physical principles. We then introduce the **basic concepts and classification** of differential equations, providing the

essential language and framework for the study that follows. Next, we examine **direction fields** – a geometric tool that allows us to visualize solutions without actually solving the equation. Finally, we discuss **isoclines**, which provide a systematic method for sketching direction fields and understanding the qualitative behavior of solutions.

1.1 Applied Problems Leading to Differential Equations

In order to apply mathematical methods to a physical or "real life" problem, we must formulate the problem in mathematical terms; that is, we must construct a **mathematical model** for the problem. Many physical problems concern relationships between changing quantities. Since rates of change are represented mathematically by derivatives, mathematical models often involve equations relating an unknown function and one or more of its derivatives. Such equations are **differential equations**.

The mathematical model for an applied problem is almost always simpler than the actual situation being studied, since simplifying assumptions are usually required to obtain a mathematical problem that can be solved. For example, in modeling the motion of a falling object, we might neglect air resistance and the gravitational pull of celestial bodies other than Earth, or in modeling population growth we might assume that the population grows continuously rather than in discrete steps.

A good mathematical model has two important properties:

1. It is sufficiently simple so that the mathematical problem can be solved.

2. It represents the actual situation sufficiently well so that the solution to the mathematical problem predicts the outcome of the real problem to within a useful degree of accuracy.

If results predicted by the model do not agree with physical observations, the underlying assumptions of the model must be revised until satisfactory agreement is obtained.

The behavior of the natural world is governed by fundamental principles and physical laws. These laws describe a wide variety of processes, including:

- The motion of fluids;
- The flow of current in electric circuits;
- The dissipation of heat in solid objects;
- The propagation and detection of seismic waves;
- The increase or decrease of populations;
- and many other phenomena.

In each case, there exists a connection between the characteristics of the phenomenon and the rates of their change relative to other variables. This connection leads to **equations containing derivatives of unknown functions** – that is, to **differential equations**.

The subject of differential equations has a rich history that dates back to the development of calculus in the 17th century. The most famous book, *Mathematical Principles of Natural Philosophy* (1687) by Isaac Newton, laid the foundation

for classical mechanics and, implicitly, the use of differential equations to describe the motion of bodies.

The term “**differential equation**” was introduced in 1676. Gottfried Wilhelm Leibniz arrived at the fundamental results of calculus independently, although a little later than Newton, but was the first to publish them. Leibniz also introduced the notation $\frac{dy}{dx}$ that we still use today.

The development of differential equations has been driven by the needs of physics, astronomy, and engineering, and continues to be a central tool in modern science and technology.

Example 1: A Falling Object

An object with mass m falls under the action of the force of gravity mg (where g is the acceleration due to gravity, 9.8 m/s^2) and a drag force $F_d = -kv$, proportional to its velocity v , where k is the drag coefficient in kg/s .

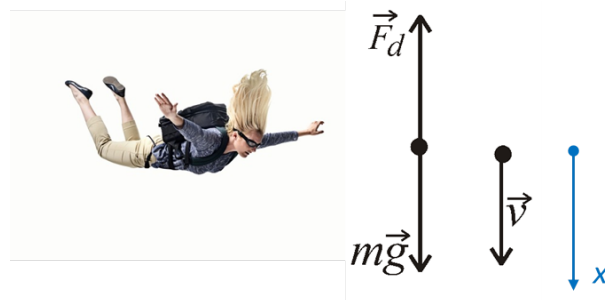


Fig. 1.1. Free-body diagram of the forces on a falling object

According to Newton’s second law of motion, the sum of forces acting on the object equals the product of mass and acceleration:

$$m \frac{dv}{dt} = mg - kv. \quad (1.1)$$

This is a first order ordinary differential equation. It can be rewritten as:

$$\frac{dv}{dt} = g - \frac{k}{m}v. \quad (1.2)$$

Its solution describes the velocity as a function of time:

$$v(t) = \frac{mg}{k} + C \exp\left(-\frac{k}{m}t\right), \quad (1.3)$$

where C is determined by the initial condition $v(0) = v_0$. As $t \rightarrow \infty$, the velocity approaches the terminal velocity:

$$\lim_{t \rightarrow \infty} v(t) = v_{\max} = \frac{mg}{k}. \quad (1.4)$$

It can be verified by substitution that this function satisfies the differential equation.

Example 2: Population Dynamics

The first population dynamics model was proposed by Thomas Malthus in 1778 in his work “*An Essay on the Principle of Population*”.

Conceptual problem statement:

- The object of study is a certain population of organisms;
- There are no limiting factors for population growth;
- The rate of increase of the population size is directly proportional to the size of the population itself.

Let $N(t)$ denote the population size at time t . The rate of change of the population, $\frac{dN}{dt}$, is proportional to the current population size $N(t)$. This can be

written as:

$$\frac{dN}{dt} = (\alpha - \beta)N, \quad (1.5)$$

where α is the birth rate coefficient; β is the death rate coefficient.

If we assume that α and β are constant, we introduce the net growth rate $r = \alpha - \beta$, and the equation simplifies to:

$$\frac{dN}{dt} = rN. \quad (1.6)$$

This is the **Malthusian model**. It is a first-order linear homogeneous ODE. Its solution is obtained by separation of variables:

$$N(t) = N_0 \exp(r(t - t_0)), \quad (1.7)$$

where $N_0 = N(t_0)$ is the initial population size.

If $r > 0$, the population grows exponentially without bound. If $r < 0$, the population decays exponentially to zero. If $r = 0$, the population remains constant.

This model applies to populations such as:

- hares in the absence of predators;
- people in a given country;
- bacteria in a nutrient medium.

However, the Malthusian model has limitations. In reality, resources are finite, and population growth is eventually limited by factors such as food supply, space, and disease. More realistic models, such as the logistic growth model, will be discussed later in the course.

Example 3: Model of a Cooling Cup of Coffee

Consider a container of a given volume filled with a liquid. The object is heated to a temperature T_0 ($\sim 95^\circ\text{C}$), and then allowed to cool down to the ambient (environment) temperature T_a . A practical question arises: How long after boiling can tea or coffee be drunk? Typically, this corresponds to a temperature of approximately 60°C .

Physical Model and Assumptions. The cooling process is governed by the principle of heat transfer. The amount of heat dQ transferred from the object to the environment in a small time interval dt is given by Newton's law of cooling:

$$dQ = \alpha S(T - T_a)dt, \quad (1.8)$$

where dQ is the amount of heat transferred in Joules (J); dt is the time interval in seconds (s); T_a is the ambient temperature in Kelvin (K); T is the current temperature of the object in Kelvin (K); S is the surface area of the object in m^2 ; α is the heat transfer coefficient in $\text{W}/(\text{m}^2 \cdot \text{K})$.

We consider a simplified case where there are no internal heat sources in the object, and the entire body has a uniform temperature. This is valid when the thermal conductivity is sufficiently high, the body dimensions are small, and the heat transfer is not excessively intense. From thermodynamics, the change in internal energy of the system is related to the temperature change by:

$$dQ = c\rho V dT, \quad (1.9)$$

where c is the specific heat capacity in $\text{J}/(\text{kg} \cdot \text{K})$; ρ is the density in kg/m^3 ; V is the volume of the body in m^3 .

Differential Equation for Cooling. Equating the heat loss from the body to the change in its internal energy, and using Newton's law of cooling, we obtain:

$$c\rho V \frac{dT}{dt} = -\alpha S(T - T_a). \quad (1.10)$$

Dividing both sides by $c\rho V$, we obtain the first-order linear ODE:

$$\frac{dT}{dt} = -r(T - T_a), \quad (1.11)$$

where

$$r = \frac{\alpha S}{c\rho V}. \quad (1.12)$$

The parameter $r > 0$ is the cooling rate constant, which depends on the physical properties of the object and the heat transfer conditions.

1.2 Basic Concepts and Classification of Differential Equations

Ordinary and Partial Differential Equations

Differential equations are classified according to the number of independent variables involved.

An **ordinary differential equation (ODE)** is an equation in which the unknown function depends on a single independent variable. For example, the equation describing the charge $Q(t)$ on a capacitor in an RLC circuit:

$$L \frac{d^2 Q}{dt^2} + R \frac{dQ}{dt} + \frac{1}{C} Q = E(t), \quad (1.13)$$

where L is inductance, R is resistance, C is capacitance, and $E(t)$ is the impressed voltage.

A **partial differential equation (PDE)** is an equation in which the unknown function depends on several independent variables. For example, the heat equation in one spatial dimension:

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}, \quad (1.14)$$

where $u(x, t)$ is the temperature distribution in a rod, and α is the thermal diffusivity.

Order of a Differential Equation

The **order** of a differential equation is the order of the highest derivative that appears in the equation.

A general n -th order ODE can be written as:

$$F\left(t, x, \frac{dx}{dt}, \dots, \frac{d^n x}{dt^n}\right) = 0. \quad (1.15)$$

Example. The Duffing equation,

$$\ddot{x} + kx + x^3 = B \cos t, \quad (1.16)$$

is a second-order ODE, since the highest derivative is $\ddot{x} = d^2x/dt^2$.

A first-order ODE, solved for the derivative (explicit form), is written as:

$$\frac{dx}{dt} = f(t, x). \quad (1.17)$$

Systems of Differential Equations

A **system of differential equations** involves several unknown functions and

their derivatives. A system of n first-order ODEs in normal form is written as:

$$\frac{dx_i}{dt} = f_i(t, x_1, x_2, \dots, x_n), \quad i = 1, 2, \dots, n. \quad (1.18)$$

In vector form, this system can be written compactly as:

$$\frac{d\mathbf{X}}{dt} = \mathbf{F}(t, \mathbf{X}), \quad (1.19)$$

where:

$$\mathbf{X}(t) = (x_1(t), x_2(t), \dots, x_n(t))^T, \quad (1.20)$$

and

$$\mathbf{F}(t, \mathbf{X}) = (f_1(t, \mathbf{X}), f_2(t, \mathbf{X}), \dots, f_n(t, \mathbf{X}))^T. \quad (1.21)$$

If the right-hand sides do not explicitly depend on t , the system is called **autonomous**.

Example. The Lotka–Volterra (predator–prey) equations,

$$\begin{cases} \frac{dy_1}{dt} = ay_1 - by_1y_2, \\ \frac{dy_2}{dt} = cy_1y_2 - dy_2, \end{cases} \quad (1.22)$$

form a system of two first-order ODEs, where y_1 is the prey population and y_2 is the predator population.

Linear and Nonlinear ODEs

An ODE is called **linear** if it can be written in the form:

$$a_n(t) \frac{d^n x}{dt^n} + a_{n-1}(t) \frac{d^{n-1} x}{dt^{n-1}} + \dots + a_1(t) \frac{dx}{dt} + a_0(t)x = g(t), \quad (1.23)$$

where the coefficients $a_i(t)$ and $g(t)$ depend only on the independent variable t , not on x or its derivatives.

If an ODE is not linear, it is called **nonlinear**.

Example. The Duffing equation,

$$\ddot{x} + kx + x^3 = B \cos t, \quad (1.24)$$

is nonlinear due to the term x^3 .

Example. The Lotka-Volterra system,

$$\begin{cases} \frac{dy_1}{dt} = ay_1 - by_1y_2, \\ \frac{dy_2}{dt} = cy_1y_2 - dy_2, \end{cases} \quad (1.25)$$

is nonlinear because of the product terms y_1y_2 .

Example. An epidemic spread model can be written as:

$$\begin{cases} \frac{dN_i}{dt} = \alpha N_i N_s - \beta N_i, \\ \frac{dN_s}{dt} = \gamma(N_0 - N_s) - \alpha N_s N_i, \end{cases} \quad (1.26)$$

where N_i is the number of infected individuals and N_s is the number of susceptible individuals.

Homogeneous and Nonhomogeneous ODEs

A linear ODE is called **homogeneous** if every term is a function of the dependent variable and its derivatives, and there is no term independent of the

unknown function and its derivatives:

$$a_n(t) \frac{d^n x}{dt^n} + a_{n-1}(t) \frac{d^{n-1} x}{dt^{n-1}} + \cdots + a_0(t)x = 0. \quad (1.27)$$

A linear ODE is called **nonhomogeneous** if it includes an additional function $g(t)$ independent of the unknown function:

$$a_n(t) \frac{d^n x}{dt^n} + a_{n-1}(t) \frac{d^{n-1} x}{dt^{n-1}} + \cdots + a_0(t)x = g(t), \quad (1.28)$$

where $g(t) \neq 0$. The trivial solution $x \equiv 0$ is a solution only of the homogeneous equation.

Example. The Duffing equation with forcing,

$$\ddot{x} + kx + x^3 = B \cos t, \quad (1.29)$$

is nonhomogeneous.

Example. Newton's law of cooling,

$$\frac{dT}{dt} = -r(T - T_a), \quad (1.30)$$

is nonhomogeneous due to the constant term rT_a .

Example. The RLC circuit equation,

$$L \frac{d^2 Q}{dt^2} + R \frac{dQ}{dt} + \frac{1}{C} Q = E(t), \quad (1.31)$$

is nonhomogeneous if $E(t) \neq 0$.

Example. The Lotka-Volterra system,

$$\begin{cases} \frac{dy_1}{dt} = ay_1 - by_1y_2, \\ \frac{dy_2}{dt} = cy_1y_2 - dy_2, \end{cases} \quad (1.32)$$

is homogeneous in the sense that there is no external forcing term.

1.3 Direction fields for first order equations

Geometric Interpretation of a First-Order ODE

Consider a first-order ordinary differential equation solved for the derivative:

$$\frac{dx}{dt} = f(t, x). \quad (1.33)$$

From calculus, the derivative dx/dt represents the slope of the tangent line to the graph of the function $x(t)$. Therefore, equation (1.33) establishes a relationship between the coordinates of a point (t, x) and the slope of the tangent to the solution curve at that point.

Key idea: For any point (t^*, x^*) in the domain, we can compute the value $f(t^*, x^*)$ and thereby determine the slope of the solution curve passing through that point, *without actually solving the differential equation*.

This shifts the problem from the analytical domain to the **geometric** one: instead of finding an explicit formula for the solution, we seek curves whose tangent directions at every point match the prescribed slopes.

Definition of a Direction Field

A **direction field** (also called a slope field) for the ODE (1.33) is a collection

of short line segments (or arrows) drawn at grid points (t_i, x_j) in the (t, x) -plane, such that the slope of the segment at (t_i, x_j) equals $f(t_i, x_j)$.

Properties:

- All segments have the same fixed length; only their *angle* varies.
- The direction field shows the *local* behaviour of solutions but does not give a single trajectory.
- Under standard existence and uniqueness conditions, exactly one integral curve passes through each point of the domain.

Constructing a Direction Field: Step-by-Step Algorithm

1. Choose a region in the (t, x) -plane.
2. Select a grid of points.
3. Choose a fixed length L for all segments.
4. At each grid point (t_i, x_j) , compute the slope:

$$k_{ij} = f(t_i, x_j).$$

5. Through (t_i, x_j) , draw the line:

$$x - x_j = k_{ij}(t - t_i).$$

6. On this line, take a segment of length L centred at (t_i, x_j) .
7. Repeat for all grid points.

Important: The length L must be smaller than the grid spacing Δt and Δx so that the segments do not overlap or cross each other.

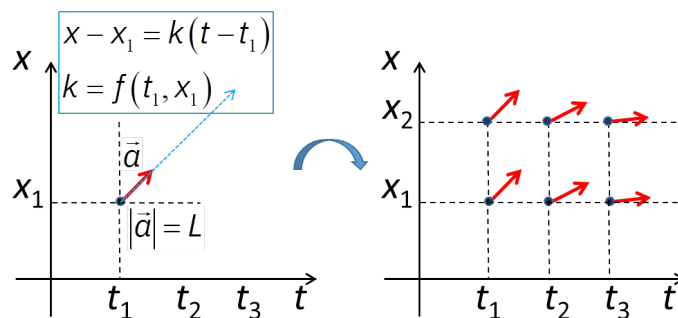


Fig. 1.2. Procedure for constructing a direction field.

Example: For $\frac{dx}{dt} = 1$, we have $k_{ij} = 1$ everywhere. Choose $L = 0.5$. At each grid point, draw a segment of length 0.5 with slope 1. All segments are parallel and have the same length.

Remark: In practice, direction fields are generated using computational tools (MATLAB, Python, Maple, etc.) because manual calculations for a large number of points are time-consuming. However, understanding the algorithm is essential for visual interpretation.

Example. Consider the elementary equation:

$$\frac{dx}{dt} = 1. \quad (1.34)$$

Here, the right-hand side is constant and does not depend on either t or x . Therefore, at *every* point (t, x) , the slope of the direction field is the same: $k = f(t, x) = 1$. All line segments are parallel and have the same slope 1. The integral curves are straight lines: $x(t) = t + C$, where C is an arbitrary constant.

The direction field shows a family of parallel lines. The constant C simply shifts the line vertically, while the slope remains unchanged everywhere. This

example demonstrates that the direction field completely determines the family of solutions; the initial condition selects one specific curve from this family.

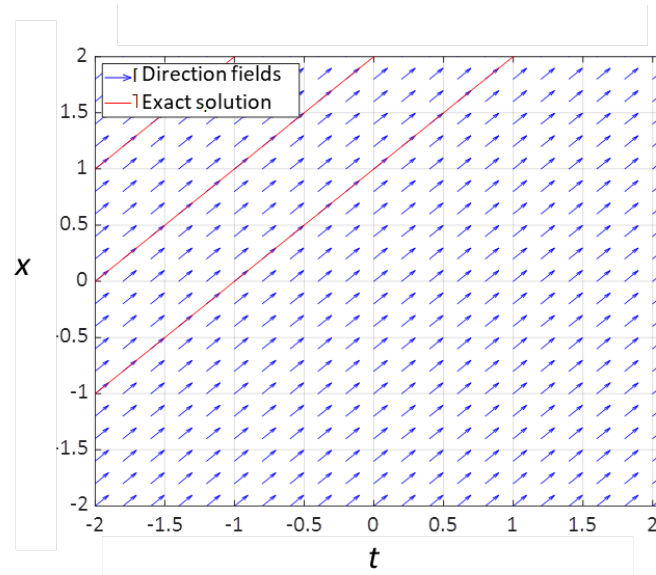


Fig. 1.3. Direction field.

Example. A Falling Object

Consider the ODE modelling the velocity of a falling object with air resistance:

$$\frac{dv}{dt} = 9.8 - \frac{v}{5}, \quad (1.35)$$

where v is the velocity in m/s. This equation comes from $m = 10$ kg, $g = 9.8$ m/s², and drag coefficient $k = 2$ kg/s.

Analysis using the direction field:

- If $v = 40$, then $\frac{dv}{dt} = 9.8 - 8 = 1.8 > 0$ – velocity increases (positive slopes).
- If $v = 60$, then $\frac{dv}{dt} = 9.8 - 12 = -2.2 < 0$ – velocity decreases (negative slopes).

- If $v = 49$, then $\frac{dv}{dt} = 0$ – horizontal segments.

The value $v = 49$ m/s is called an **equilibrium solution** (or steady-state solution). The direction field shows that all solutions tend to this value as $t \rightarrow \infty$, regardless of the initial condition.

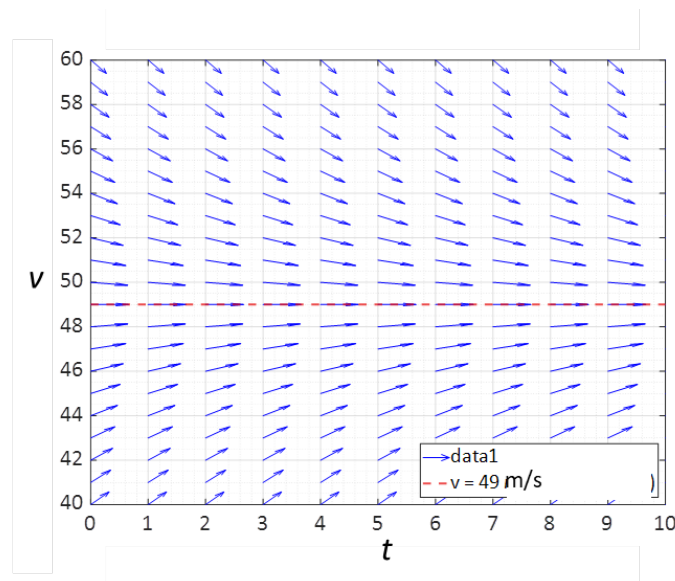


Fig. 1.4. Direction field.

Example.

Consider

$$\frac{dy}{dx} = \frac{y}{x}. \quad (1.36)$$

At every point (x, y) (except the origin), the slope equals y/x , which is exactly the slope of the straight line from the origin to that point. Hence, the integral curves are straight lines $y = Cx$, since their direction coincides everywhere with the direction field.

This example illustrates that the direction field can *suggest* the form of the solution even without analytical integration.

Advantages and Limitations of Direction Fields

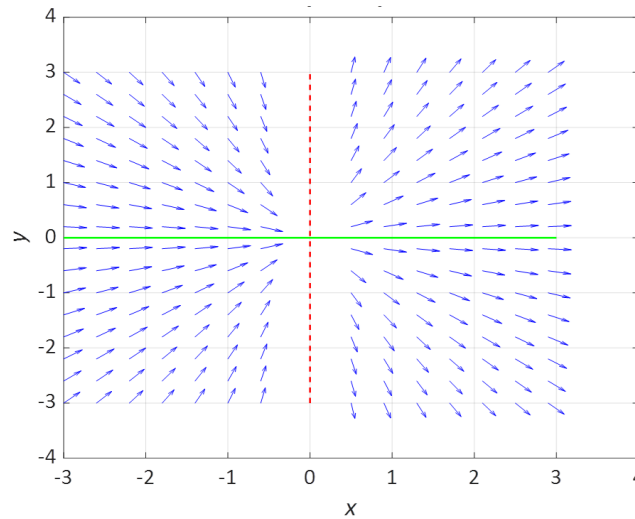


Fig. 1.5. Direction field.

Direction fields are a powerful visual tool, but they have both strengths and weaknesses.

Advantages:

- Requires no analytical solution.
- Visually shows solution behaviour.
- Helps detect equilibrium solutions.
- Works for any (even nonlinear) equations.
- Easy to understand and interpret.

Limitations:

- Gives only qualitative information.
- Accuracy depends on grid density.
- Time-consuming for manual plotting.

- Does not provide numerical values at specific points.
- Can be misleading if the grid is too coarse.

1.4 Isoclines

Definition and Motivation

Constructing a direction field requires evaluating $f(t, x)$ at every grid point, which can be expensive. A more economical approach is to draw **curves of constant slope** and then place direction segments along these curves.

An **isocline** (from Greek *isos* – equal, and *klinein* – to slope) for the ODE

$$\frac{dx}{dt} = f(t, x) \quad (1.37)$$

is a curve defined by

$$f(t, x) = C, \quad (1.38)$$

where C is a constant. Along an isocline, the slope of the direction field is the same everywhere.

Geometric Solution Algorithm Using Isoclines

1. **Define the ODE:** Start with $\frac{dx}{dt} = f(t, x)$.
2. **Select slope values:** Choose several constants $C_1, C_2, \dots, C_n$ (e.g., $-2, -1, 0, 1, 2$).
3. **Plot isoclines:** For each C_i , plot the curve $f(t, x) = C_i$.
4. **Draw direction segments:** Along each isocline, draw short segments with slope C_i .

5. Construct integral curves:

- Start from the initial point (t_0, x_0) .
- Follow the direction segments from one isocline to the next.
- Maintain tangency to the segments.
- Draw a smooth curve connecting the segments.

6. Refine: Use more isoclines for higher accuracy.

Example

Let's us consider the following example:

$$\frac{dx}{dt} = \sqrt{x^2 + t^2}. \quad (1.39)$$

The isoclines are given by

$$\sqrt{x^2 + t^2} = C \implies x^2 + t^2 = C^2. \quad (1.40)$$

These are **circles** centred at the origin with radius C . Along each circle, the slope is constant and equals C : the isocline degenerates to a single point $(0, 0)$, where $f(t, x) = 0$. Note that the ODE is not differentiable at this point.

For example:

- $C = 0$: degenerate circle (the origin), slope 0.
- $C = 1$: circle of radius 1, all segments have slope 1 (45°).
- $C = 2$: circle of radius 2, all segments have slope 2 ($\approx 63^\circ$).

This makes it very easy to visualise the direction field without computing the square root at hundreds of points.

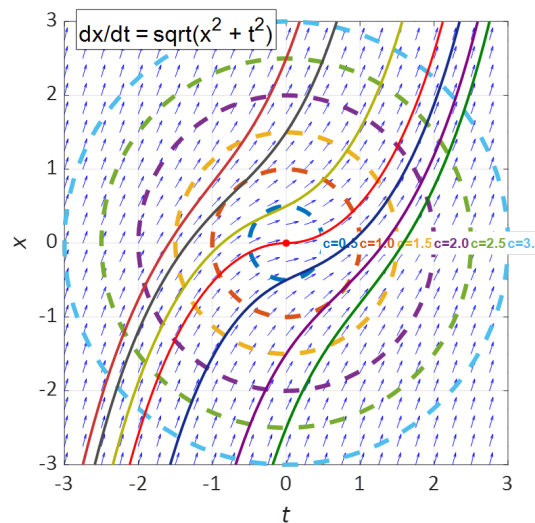


Fig. 1.6. Isoclines

Closing Remark

The ability to construct direction fields and isoclines is the first step toward **geometric thinking** in the theory of ordinary differential equations. It develops intuition that later helps in:

- selecting appropriate solution methods,
- analysing stability,
- interpreting results of mathematical models.

Even without solving an ODE analytically, direction fields and isoclines allow us to *see* the behaviour of solutions.